

Draft VCS Methodology

VM0042

IMPROVED AGRICULTURAL LAND MANAGEMENT

Draft Version 3.0

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Sectoral Scope 15: Agriculture

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1 SUMMARY DESCRIPTION

Table 1-1. Additionality and crediting baseline methods

Additionality, Crediting Method and Mitigation Outcome	
Additionality	Project Method
Crediting Baseline	Project Method
Mitigation Outcome	Reductions and Removals

This agricultural land management (ALM) methodology provides procedures to estimate the greenhouse gas (GHG) emission reductions (CO₂, CH₄, and N₂O) and carbon dioxide removals resulting from the adoption of improved ALM practices. The methodology is compatible with regenerative agriculture and has a particular focus on increasing soil organic carbon (SOC) storage.

The crediting baseline and additionality are determined via a project method (Table 1-1). The baseline scenario is determined as the continuation of baseline ALM practices. Practices in the baseline scenario are determined by applying a minimum three-year historical look-back period to produce an annual schedule of activities (i.e., tillage, planting, harvest, and fertilization events) for each management unit within the project area (e.g., for each field), to be repeated over the baseline period.¹ Baseline emissions/stock changes are then modeled. Alternatively, baseline SOC stock change may be directly measured in “baseline control sites” managed according to baseline practices as set out in the schedule of activities. The baseline scenario is reassessed as required by the most recent version of the *VCS Standard*, and revised, where necessary, to reflect current agricultural production practices in the region.

Additionality is demonstrated by a barrier analysis and showing that the practice change implemented under the project activity is not common practice. A practice change constitutes any of the following:

- Adoption of a new practice (e.g., adoption of one or more of the practices covered in the categories included in Applicability Condition 1);
- Cessation of a pre-existing practice (e.g., implementing no-tillage or avoidance of residue burning);
- Adjustment to a pre-existing practice (e.g., optimize fertilizer application or improve grazing management); or

¹ Project proponents must reassess ALM project baselines every ten years or every five years, where regional practice changes are more frequent and/or data becomes available to update the baseline. See the most recent version of the *VCS Standard* for further details on baseline reassessment requirements.

- Some combination of the above.

Any quantitative adjustment (e.g., decrease in fertilizer application rate) must exceed 5% of the average pre-existing value in the historical look-back period to qualify as a practice change.

The methodology provides three approaches to quantifying GHG emission reductions and carbon dioxide (reductions and removals) resulting from the adoption of improved ALM practices.

Quantification Approach 1: Measure and Model (QA1)

A biogeochemical, process-based model is used to quantify SOC stock changes, CH₄ fluxes from soil methanogenesis, and N₂O fluxes from the use of nitrogen fertilizers and the use of nitrogen-fixing species, and manure deposition. Edaphic characteristics and actual agricultural practices implemented, measured initial SOC stocks, and climatic conditions in sample fields are used as model inputs. Periodic measurements of SOC stocks are required every five years at minimum (see Section 8). *VM0053 Model Calibration, Validation, and Uncertainty Guidance for Biogeochemical Modeling for Agricultural Land Management Projects* establishes the requirements for calibration and validation of biogeochemical models.

Quantification Approach 2: Measure and Re-Measure (QA2)

Direct measurement is used to quantify SOC stock changes, and woody biomass carbon stock changes. This approach directly measures SOC carbon stock changes in the baseline scenario in linked baseline control sites. It is the only eligible approach for woody biomass carbon stock change quantification, where baseline woody biomass carbon stock change cannot be set to zero per the conditions defined in Section 6.3.

Quantification Approach 3: Default Factors (QA3)

CO₂ flux from fossil fuel combustion and N₂O and CH₄ fluxes are calculated using default emission factors.

The quantification approach varies by emission/removal type. Approaches to quantification of contributing sources of CO₂, CH₄, and N₂O emissions are described in Section 8.

Project proponents must use the templates available on the VM0042 webpage to prepare the documents required for project validation and verification.

2 SOURCES

This methodology uses the following methodologies, modules, and tools:

- *VM0032 Sustainable Management of Native Grasslands through Adjustment of Fire and Grazing*
- *VM0047 Afforestation, Reforestation, and Revegetation*
- *VM0051 Improved Management in Rice Production Systems*

- *VMD0053 Model Calibration, Validation, and Uncertainty Guidance for Biogeochemical Modeling for Agricultural Land Management Projects*
- *VMD0054 Module for Estimating Leakage from ARR Activities*
- *VT0008 Additionality Assessment*
- *VT0009 Combined Baseline and Additionality Assessment*
- *VT0014 Estimating Organic Carbon Stocks Using Digital Soil Mapping*
- *A/R Methodological Tool: Tool for the Identification of Degraded or Degrading Lands for Consideration in Implementing CDM A/R Project Activities²*
- *TOOL16 Methodological Tool: Project and Leakage Emissions from Biomass*

3 DEFINITIONS

In addition to the definitions set out in the VCS document *Program Definitions*, the following definitions apply to this methodology.

Baseline control site

Defined area that is managed according to baseline (pre-project) practices (as set out in the schedule of activities) for direct measurement of baseline soil organic carbon and, if applicable, woody biomass carbon stock change. It is linked to and representative of the land in one or more quantification units. The area of baseline control sites does not count as part of the project area. Baseline control sites are only used under Quantification Approach 2.

Crop functional group

Group of crop species and varieties with unique combination of the following attributes: 1) Photosynthetic pathway (C3/C4/CAM); 2) Annual (and biennial) or perennial (A/P); 3) N fixation 4) Growth form (herbaceous/woody/succulent); 5) flooded/non-flooded (Y/N). The full list is provided in Appendix 8.

Cultivated grassland

Intensively managed pastures and grasslands through tillage, irrigation, and application of synthetic fertilizers.

Historical look-back period

The time period prior to the initial crediting period start date covering at minimum three years and one complete crop rotation. The historical look-back period is used to determine the baseline scenario and produce the schedule of activities (see definition below).

² This tool was deactivated by the CDM as it is no longer required as a standalone document (see CDM Executive Board Meeting Report 75 from 30 September–4 October 2013). There were no technical concerns with the procedures described in the tool; therefore, it is still valid in the context of VM0042.

Improved agricultural land management practice

An agricultural practice yielding increased soil organic carbon storage or further reductions and removals, that involves a refinement to fertilizer or other soil amendment application, water management/irrigation, tillage, residue management, crop planting and harvesting, and/or grazing practices. Proponents may implement an individual practice or a combined (stacked) practice on each field or management unit part of their project.

Inelastic neutron scattering (INS)

An in-field (in situ) measurement technique based on the detection and analysis of gamma rays emitted by soil elements after irradiation with neutrons. It is also known as neutron-stimulated gamma ray analysis or spectroscopy.

Infrared spectroscopy

Mid-infrared (MIR), near-infrared (NIR), and visible near-infrared (Vis-NIR) spectroscopy, including diffuse reflectance spectroscopy (DRS) and diffuse reflectance infrared Fourier transform spectroscopy (DRIFT). Vis-NIR combines the visible and near-infrared electromagnetic range and usually refers to a wavelength range from 350 to 2500 nm (visible range is between 350 and 700 nm). MIR covers the range between 4000 cm^{-1} and 600 (or 400) cm^{-1} , depending on the instrument.

Laser-induced breakdown spectroscopy (LIBS)

Application of a high-energy pulse to soil samples to generate a high-temperature plasma, which emits radiation at different wavelengths depending on the elements present in the sample

Management unit

The smallest functional unit of agricultural land at which homogenous management decisions are made, and most granular management activity data is available (e.g., crop field, pasture, paddock, grazing cell). The sum of the area of all management units implementing project activities in a given monitoring period is equal to the project area. A management unit is equivalent to a project activity instance (instance), in the context of this methodology.

Nitrogen-fixing species

Any plant species that associates with nitrogen-fixing microbes found within nodules formed on the roots, including but not limited to soybeans, alfalfa, and peas

Organic nitrogen fertilizer

Any organic material containing nitrogen, including but not limited to animal manure, compost, and biosolids

Professional agronomist

An individual with demonstrated specialized knowledge, skills, education, experience, or training in crop and/or soil science. Such individuals may be agricultural experts like soil

scientists, husbandry specialists, agronomists, or representatives of a governmental agricultural body.

Project domain

Set of conditions (including crop type, soil texture, and climate) in which biogeochemical model application has been validated (see VCS Module *VMD0053 Model Calibration, Validation, and Uncertainty Guidance for Biogeochemical Modeling for Agricultural Land Management Project*).

Quantification unit

A geographically defined area within the project (subset of the project area) defined by batches of project activity instances for which reductions and removals are quantified using the selected quantification approach with their respective uncertainty deductions. All project activity instances included within a quantification unit have a common enrollment year and historical look-back period.

Rehabilitation

Process supported by improved ALM activities that recover and enhance the functionality, productivity, and services of degraded agricultural productive systems, without necessarily returning them to their natural ecological composition, function, and structure.

Sampling unit

An individual point or an area (i.e., management unit) eligible for sampling. The selected sampling units are those actually chosen for sampling according to the sampling design.

Schedule of activities

Annual schedule of historical management practices applied (e.g., tillage, planting, harvest, grazing, irrigation, and fertilization events) and crop functional groups cultivated (for cropland) in the baseline scenario derived from the historical look-back period. These practices are determined following the data requirements given in Box 1.

SOC Stratum (strata)

SOC Strata refers to subdivisions of a given area that group soils with similar characteristics influencing SOC stocks and their temporal dynamics—such as soil type, land use, management practice, climate, topography, and baseline SOC levels—so that variability within each group is minimized and differences between groups are maximized. The SOC stocks within one stratum are more homogenous than the SOC stocks of different strata. SOC stratification (i.e., the process of defining strata) aims to improve statistical power for detecting changes over time.

Synthetic nitrogen fertilizer

Any fertilizer containing nitrogen made by chemical synthesis (solid, liquid, gaseous) or electrochemical processes. This may be a single nutrient fertilizer product (only including N), or any other synthetic fertilizer containing N, such as multi-nutrient fertilizers (e.g., N–P–K fertilizers) and “enhanced-efficiency” N fertilizers (e.g., slow release, controlled release, and stabilized N fertilizers).

Woody biomass

Biomass in plants with hard, lignified stems, for example, trees, shrubs, palms and bamboo.

4 APPLICABILITY CONDITIONS

This methodology applies to a broad range of project activities that increase SOC storage, increase woody biomass carbon stocks and/or decrease net emissions of CO₂, CH₄, and N₂O from ALM operations compared to the baseline scenario. The methodology is globally applicable.

This methodology is applicable under the following conditions:

- 1) Projects introduce or implement one or more new changes to pre-existing ALM practices which:
 - a) Improve fertilizer (organic or inorganic) management;
 - b) Improve water management;
 - c) Improve irrigation;
 - d) Reduce tillage
 - e) Improve residue management;
 - f) Improve crop planting and harvesting (e.g., agroforestry, crop rotations, cover crops); and/or
 - g) Improve grazing practices.

Appendix 1 provides a non-exhaustive list of eligible ALM practices. A change in practice constitutes adoption of a new practice, cessation of a pre-existing practice, or adjustment to a pre-existing practice that results in reductions or removals.

- 2) Projects introduce or implement quantitative adjustments (e.g., decrease in synthetic fertilizer application rate) that exceed 5% of the pre-existing value, calculated as the crop- or grassland-specific average value over the historical look-back period, developed for the baseline schedule of activities (see Section 6). Appendix 1 gives additional details and guidance on practices and on determining practice change.
- 3) Project activities are implemented on land that is either cropland or cultivated grassland at the initial crediting period start date. Cropland under woody perennial crops³ (e.g., orchards) or agroforestry (e.g., shaded coffee) is permitted even where the

³ Woody perennial crops are trees and shrubs having a life cycle lasting more than two years, excluding cultivated annual species with lignified tissues, such as cotton or hemp.

canopy cover may exceed national definitions of a forest, provided that the land is under agricultural production at the initial crediting period start date and remains under agricultural production.⁴

The land must remain cropland or cultivated grassland throughout the project lifetime except for land rehabilitation purposes under the following scenarios:

- a) Introduction of temporary grassland into cropland where it is demonstrated, prior to the initial crediting period start date and to the addition of new project activity instances, that the integration of forage crops (e.g., annual/perennial grasses, legumes) into annual crops is part of a planned, long-term ALM system (e.g., integrated crop-livestock system). Project proponents must provide documentation of the long-term management plans, covering the duration of the project, that describe proposed practices, crops, and expected benefits and outcomes of integrated grassland-cropland management; or
- b) A one-time ecosystem conversion from cultivated grassland to cropland or vice versa where it is demonstrated, prior to the project start date, that:
 - i) project lands in the baseline scenario are in a state of high ecosystem degradation, according to the most recent version of the *VCS Standard*,
 - ii) the introduction of improved land use change practices would significantly improve soil health.
 - iii) lands in a state of high ecosystem degradation at the project start date will continue in the baseline scenario due to the presence of degradation drivers or pressures. See Appendix 2 for procedures on how to propose this type of land use change.

This methodology is not applicable under the following conditions:

- 4) The project area has been cleared of native ecosystems within the 10-year period immediately prior to the initial crediting period start date.
- 5) The project activity is expected to cause a sustained reduction in productivity of greater than 5%, as demonstrated by peer-reviewed and/or published studies on activity in the region or a comparable region.
- 6) The project activities include biochar application. Biochar may be applied as a soil amendment in the project area provided that the total organic carbon content⁵ of the biochar applied is subtracted from the estimated SOC stock change in the project scenario at each verification event.

⁴ In this context, agricultural production means cultivating crop plants and/or raising livestock to generate agricultural goods such as food, fuel, and fiber for human use.

⁵ To estimate the total carbon content of the applied biochar, project proponents should follow the procedures set out in Sections 8.2.2.1 or 8.2.2.2 (for high- or low-technology production facilities, respectively) in the most recent version of *VM0044 Methodology for Biochar Utilization in Soil and Non-Soil Applications*. Where the technology production facility type is not known, procedures in the low-technology approach (Section 8.2.2.2) should be followed for conservativeness.

- 7) The project activities occur on a wetland; this condition does not exclude crops subject to artificial flooding where it is demonstrated that crop cultivation does not impact the hydrology of any nearby wetlands.
- 8) The project activities occur on natural, untilled, uncultivated grasslands, savannas, or grassy woodlands, rangelands or bushlands.⁶
- 9) Project activities involve clearing, burning, or mechanical disturbance of existing vegetation (e.g., for site preparation) that reduce stocks in monitored carbon pools.
- 10) Fire management in cultivated grasslands is not an eligible project activity.

5 PROJECT BOUNDARY

Carbon pools and GHG sources included in the project boundary in the baseline and project scenarios are listed in Table 5-1 and Table 5-2, respectively. Specific carbon pools and GHG sources may be deemed *de minimis* and need not be accounted for (i.e., value set to zero) where together the omitted decrease in carbon stocks (in carbon pools) or increase in GHG emissions (from GHG sources) amounts to less than 5% of the total GHG benefit generated by the project. This includes sources and pools that cause project and leakage emissions. The project must demonstrate at each verification event that the specific carbon pools and GHG sources that were deemed *de minimis* at validation still meet the 5% requirement. This and all subsequent references to *de minimis* demonstration are determined by applying procedures in Appendix 7.

Table 5-1. Selected carbon pools in the baseline and project scenarios

Source	Included?	Justification/Explanation
Aboveground woody biomass	Yes / Optional	Aboveground woody biomass must be included where project activities significantly reduce the pool compared to the baseline. In all other cases, aboveground woody biomass is an optional pool. Note that above- and belowground woody biomass are jointly quantified.
Aboveground non-woody biomass	No	Carbon pool is not included because it is not subject to significant changes or potential changes are transient in nature.
Belowground woody biomass	Optional	Belowground woody biomass may optionally be included where project activities significantly increase the pool compared to the baseline. Note that above- and belowground woody biomass are jointly quantified.
Belowground non-woody biomass	No	Carbon pool is not included because it is not subject to significant changes or potential changes are transient in nature.

⁶ VM0032, v2.0, must be used for project activities in these agro-ecosystems.

Dead wood	No	Carbon pool is not included because it is not subject to significant changes or potential changes are transient in nature.
Litter	No	Carbon pool is not included, because it is not subject to significant changes or potential changes are transient in nature.
SOC	Yes / Mandatory	Major carbon pool affected by project activity that is expected to increase in the project scenario. The SOC pool must be included in the project boundary (i.e., it must be monitored and is not allowed to be deemed <i>de minimis</i>).
Wood products	No	Carbon pool is optional for ALM project methodologies and may be excluded from the project boundary.

Table 5-2. GHG sources and sinks accounted for as baseline, project, and leakage emissions

Source/Sink		Type	Gas	Included?	Justification/Explanation
Baseline and Project	Fossil Fuel	Source	CO ₂	S*	Must be included where fossil fuel use by vehicles (mobile sources, such as trucks, tractors) and mechanical equipment significantly increases CO ₂ emissions in the project compared to the baseline scenario.
			CH ₄	No	Not a significant source
			N ₂ O	No	Not a significant source
			Other	No	No other significant greenhouse gases
	Liming	Source	CO ₂	S*	Must be included where application of limestone or dolomite as soil amelioration represents a significant source of CO ₂ emissions in the project compared to the baseline scenario.
			CH ₄	No	Not a significant source
			N ₂ O	No	Not a significant source
			Other	No	No other significant greenhouse gases
	Soil methanogenesis	Source	CO ₂	No	Not a significant source
			CH ₄	S*	Must be included where changes in levels of anoxic conditions in soils lead to significant changes in soil methanogenesis in the project compared to the baseline scenario.

	Enteric fermentation		N ₂ O	No	Not a significant source
			Other	No	No other significant greenhouse gases
		Source	CO ₂	No	Not a significant source
			CH ₄	S*	Must be included where livestock are present in the baseline or project scenarios.
			N ₂ O	No	Not a significant source
			Other	No	No other significant greenhouse gases
	Manure deposition	Source	CO ₂	No	Not a significant source
			CH ₄	Yes	Must be included where livestock are present in the project or baseline scenarios.
			N ₂ O	Yes	Must be included where livestock are present in the project or baseline scenarios.
			Other	No	No other significant greenhouse gases
	Use of nitrogen fertilizers	Source	CO ₂	No	Not a significant source
			CH ₄	No	Not a significant source
			N ₂ O	Yes	Must be included where: - the project area would have been subject to nitrogen fertilization in the baseline scenario; or - nitrogen fertilization is greater in the project scenario than the baseline scenario.
			Other	No	No other significant greenhouse gases
	Use of nitrogen-fixing species	Source	CO ₂	No	Not a significant source
			CH ₄	No	Not a significant source
			N ₂ O	Yes	Must be included where nitrogen-fixing species are planted in the baseline or project scenario.
			Other	No	No other significant greenhouse gases
	Biomass burning	Source	CO ₂	S*	Quantified as stock change in the aboveground woody biomass pool, rather than as an emissions source (see Table 5-1).

Leakage			CH ₄	S*	Must be included where biomass burning increases in the project compared to the baseline scenario
			N ₂ O	S*	Must be included where biomass burning increases in the project compared to the baseline scenario
			Other	No	No other significant greenhouse gases
	Manufacturing of synthetic nitrogen fertilizers	Source	CO ₂	S*	May be included when an emissions-intensive product applied in the baseline (e.g., grey ammonia) is entirely replaced by a low-carbon product in the project (e.g., green ammonia),
			CH ₄	No	Not a significant source
			N ₂ O	Yes	May be included when an emissions-intensive product applied in the baseline (e.g., grey ammonia) is entirely replaced by a low-carbon product in the project (e.g., green ammonia),
			Other	No	No other significant greenhouse gases
	New application of organic amendments from outside the project area	Source	CO ₂	Yes	Must be included where new or additional manure, compost, or biosolids are applied in the project that were not applied in the historical look-back period.
			CH ₄	No	Not a significant source
			N ₂ O	No	Not a significant source
			Other	No	No other significant greenhouse gases
	Livestock displacement	Source	CO ₂	Yes	Must be included when livestock production declines unless demonstrated that livestock was not displaced. Must be quantified applying VMD0054.
			CH ₄	Yes	Must be included when livestock production declines unless demonstrated that livestock was not displaced. Must be quantified assuming livestock population in the baseline scenario.
			N ₂ O	Yes	Must be included when livestock production declines unless demonstrated that livestock was not displaced. Must be quantified assuming livestock population in the baseline scenario.

			Other	No	No other significant greenhouse gases
	Production declines	Source	CO ₂	Yes	Must be quantified when initial implementation of ALM practices leads to production declines in the first five years after initial crediting period start date.
			CH ₄	Yes	Not a significant source
			N ₂ O	Yes	Not a significant source
			Other	No	No other significant greenhouse gases
	Diversion of biomass residues	Source	CO ₂	Yes	Must be quantified where manure or crop residue management is a component of the project activity, and the manure or crop residues are diverted from energy applications in the baseline scenario.
			CH ₄	Yes	
			N ₂ O	Yes	
			Other	No	No other significant greenhouse gases
	Manufacturing of synthetic nitrogen fertilizers	Source	CO ₂	S*	Must be included where application rate of synthetic nitrogen fertilizers is greater in the project scenario than the baseline scenario. May be conservatively excluded where the application rate of synthetic nitrogen fertilizer is higher in the baseline than the project scenario.
			CH ₄	No	Not a significant source
			N ₂ O	S*	Must be included where the application rate of synthetic nitrogen fertilizers is greater in the project scenario than in the baseline scenario. May be conservatively excluded where the application rate of synthetic nitrogen fertilizer is higher in the baseline than the project scenario.
			Other	No	No other significant greenhouse gases

S* – Must be included where the project activity significantly increases emissions (i.e., by more than 5%) compared to the baseline scenario and may be included where the project activity reduces emissions compared to the baseline scenario.

6 BASELINE SCENARIO

This methodology uses the project method to determine the baseline scenario. Procedures to determine the baseline scenario distinguish between 1) project validation and establishing grouped project eligibility, and 2) quantifying baseline C stock changes and GHG emissions. This is because determining the baseline scenario requires a qualitative description of pre-project ALM practices, while quantifying baseline emissions requires a quantitative specification of those practices.

First, pre-project ALM practices observed across all management units are analyzed to determine a baseline scenario applicable to the project at validation. For grouped projects, this analysis also defines eligibility areas with a common baseline scenario for new management units (i.e., instances) added after initial validation. Then, for each management unit, a schedule of activities is developed describing baseline practices to be modeled, measured and/or calculated, based on the selected quantification approach.

For ex-ante ERR estimates required at project validation, the information source hierarchy defined in Box 1 must be used to collect data to estimate baseline and project emissions of initial project activity instances (i.e., management units). If aggregated project-scale estimates are provided due to lack of data availability, the project proponent must demonstrate how those estimates are accurate, conservative and applicable to the project conditions. Figure 1 provides a decision tree to further clarify the selection of qualitative and quantitative data for baseline scenario description and baseline emission quantification, respectively.

Box 1: Sources of qualitative and quantitative data

Sources of information for all undefined activity/management-related model input variables (see Table 8-1 and Table 8-2), and all parameters relevant to the baseline scenario (i.e., all parameters with the subscript *bs/* that reference Box 1 in its respective parameter table), must follow the requirements detailed below.

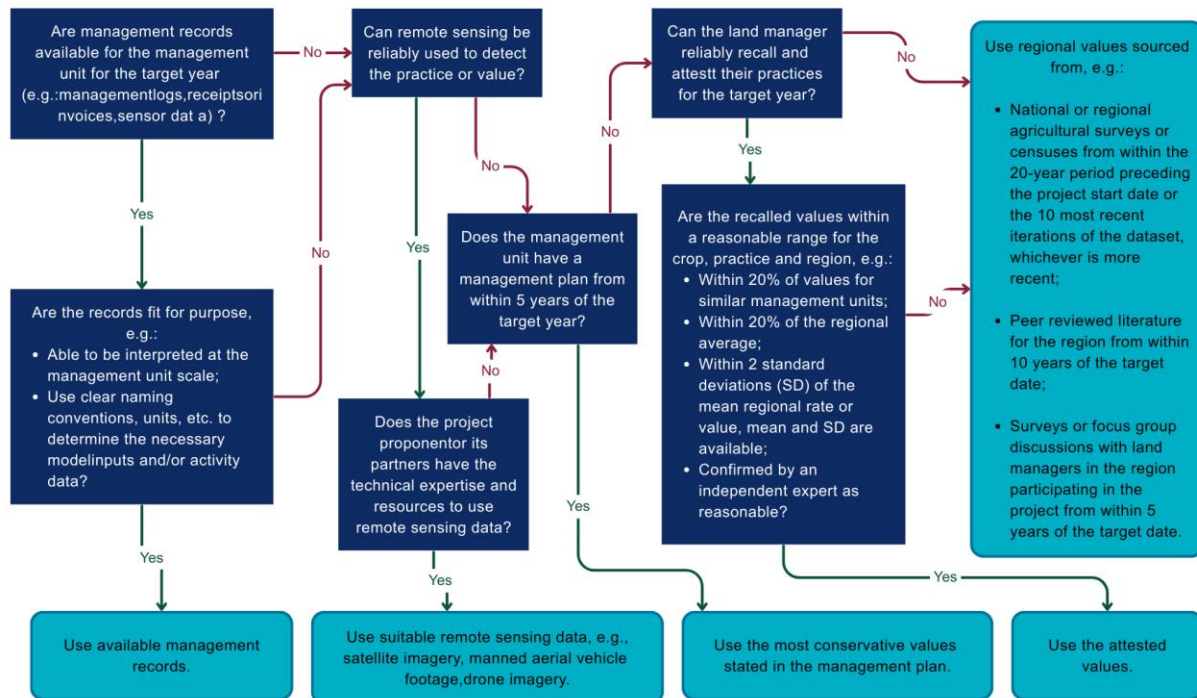
All qualitative information on ALM practices must be determined via consultation with, and substantiated with a signed attestation from, the farmer or landowner of the sample field during that period. Where the farmer or landowner is not able to provide qualitative information (e.g., a sample field is newly leased), the project proponent must follow the quantitative information hierarchy outlined below.

The following list specifies the prioritization of allowable sources of quantitative information on ALM practices. Higher-ranked sources are considered more reliable and must be prioritized. Projects must demonstrate that the selection of the source is based on data availability in the project region.

- 1) Historical management records supported by one or more forms of documented evidence pertaining to the selected sample field and period $t = -1$ to $t = -3$ (e.g., management logs, receipts or invoices, farm equipment specifications, logs or files containing machine and/or sensor data) or remote sensing (e.g., satellite imagery, manned aerial vehicle footage, drone imagery), where requisite information on ALM practices can be reliably determined with these methods (e.g., tillage status, crop type, irrigation)
- 2) Historical management plans supported by one or more forms of documented evidence pertaining to the selected sample field and period $t = -1$ to $t = -3$ (e.g., management plan, recommendations in writing solicited by the farmer or landowner from an agronomist). Where more than one value is documented in historical management plans (e.g., where a range of application rates are prescribed in written recommendations), the principle of conservativeness must be applied and the value that results in the lowest expected emissions (or highest rate of stock change) in the baseline scenario must be selected.
- 3) A signed attestation from the farmer or landowner of the sample field during that period – where the attested value does not deviate significantly from other evidence-supported values for similar fields (e.g., fertilizer data from adjacent fields with the same crop, adjacent years of the same field, government data on application rates in that area, or statement from a local extension agent regarding local application rates). In circumstances where this requirement is not met, Option 4 must be followed.
- 4) Regional (sub-national) average values derived from agricultural census data or other sources from within the 20-year period preceding the project start date or the 10 most recent iterations of the dataset, whichever is more recent. Where estimates have been disaggregated by relevant crop or ownership classes, those should be used. The estimates must be substantiated with a signed attestation from the farmer or landowner of the sample field during that period. Examples include the USDA National Agricultural Statistics Service Quick Stats database and USDA Agricultural Resource Management Survey.

This hierarchy applies to any additional quantitative inputs required by the model (Quantification Approaches 1 and 2) or default factor (Quantification Approach 3) selected. The principle of conservativeness must be applied in all cases.

Figure 1: Decision tree to determine the source of information for qualitative and quantitative data for baseline scenario description and baseline emission quantification.



6.1 Baseline Scenario Determination for Validation and Grouped Projects

The baseline scenario describes the ALM activities in the project area that would occur in the absence of the project activity. Considering current market trends, government policies and legal requirements, socioeconomic conditions, and technological developments in the agricultural sector, the continuation of pre-project ALM practices is deemed the most plausible baseline scenario (see Appendix 3 for further justification).

A historical look-back period, defined as at least three years prior to the initial crediting period start date, or a complete crop rotation, whichever is longer, is established. Pre-project qualitative ALM practices during the historical look-back period are then determined for each management unit following the specifications outlined in Table 6-1.⁷ A description summarizing the range

⁷ Note that the same historical look-back period is also used to develop a schedule of activities to determine the corresponding baseline emissions for each management unit. The difference is that baseline scenario determination requires a *qualitative description*, whereas baseline emission quantification requires *quantitative specification*, of the ALM practices in the historical look-back period. For more information on developing the schedule of activities for baseline emissions see Section 6.2, and for details on quantifying baseline emissions, refer to Section 8.3.

of variability in pre-project qualitative ALM practices observed in all initial management units within this historical look-back period is then provided – this is the baseline scenario.⁸

Table 6-1. Minimum specifications for ALM practices in the baseline scenario. Qualitative variables are used to determine the baseline scenario. Quantitative variables are used to develop a schedule of activities for the quantification of baseline emissions (see Section 6.2).

ALM Practice	Qualitative	Quantitative
Crop Planting and Harvesting	• Crop functional group(s) • Crop rotation (Y/N) • Cover crops (Y/N) • Intercropping (Y/N)	• Approximate date(s) planted • Approximate date(s) harvested/terminated • Crop yield
Nitrogen Fertilizer Application	• Organic N fertilizer (Y/N) • Organic N fertilizer (type) • Synthetic N fertilizer (Y/N) • Synthetic N fertilizer (type)	• Organic N fertilizer (manure, compost or biosolids) type application rate • N application rate in synthetic fertilizer
Tillage and/or Residue Management	• Tillage (Y/N) • Tillage intensity (type) • Crop residue removal (Y/N)	• Depth of tillage • Frequency of tillage • Percent of soil area disturbed • Percent of crop residue removed
Water Management	• Irrigation (Y/N) • Flooding (Y/N)	• Irrigation rate
Grazing Practices	• Grazing (Y/N) • Grazing system description (continuous, rotational, strip, etc.) • Animal type (where applicable) • Harvesting/mowing (Y/N)	• Animal stocking rate (i.e., number of animals and length of time grazing in each area annually) • Frequency of harvest
Liming	• Application of calcitic limestone or dolomite (Y/N)	• Calcitic limestone or dolomite application frequency • Calcitic limestone or dolomite application rate
Woody biomass management	• Chemical or manual control of competing vegetation (Y/N) • Burning of vegetation (Y/N) • Pruning (Y/N)	• Planting date(s) • Harvesting cycle per species • Frequency of control methods • Pruned biomass

Note - For dichotomous qualitative variables, 'Yes' must be selected when the ALM practice is applied at least once during the historical look-back period. When the ALM practice is not applied during any year of the historical look-back period, the variable must be defined as 'No.'

⁸ Quantitative variability as per Table 6-1 is only required for calculating baseline emissions and does not need to be specified for the baseline scenario.

The example in Table 6-2 describes the baseline scenario determined for a hypothetical project implementing the improved ALM practices of intercropping, compost application, crop residue retention, and zero tillage (i.e., the project scenario) in three initial instances.

Table 6-2. Simplified example showing how to determine the baseline scenario for a hypothetical project with three initial instances.

ALM practices	Instance 1	Instance 2	Instance 3	Baseline scenario
Crop Planting and Harvesting	• Crop functional group(s): C4-a-nN-h-nf⁹ (Maize) and C3-a-N-h-nf¹⁰ (Soybean) • Crop rotation (Y/N): Y • Cover crops (Y/N): N • Intercropping (Y/N): N	• Crop functional group(s): C4-a-nN-h-nf (Maize), C3-a-nN-h-nf (Wheat) and C3-a-N-h-nf (Soybean) • Crop rotation (Y/N): Y • Cover crops (Y/N): N • Intercropping (Y/N): N	• Crop functional group(s): C4-a-nN-h-nf (Maize) and C3-a-N-h-nf (Soybean) • Crop rotation (Y/N): Y • Cover crops (Y/N): N • Intercropping (Y/N): N	• Crop functional group(s): C4-a-nN-h-nf (Maize), C3-a-nN-h-nf (Wheat) and C3-a-N-h-nf (Soybean) • Crop rotation (Y/N): Y • Cover crops (Y/N): N • Intercropping (Y/N): N
Nitrogen Fertilizer Application	• Organic N fertilizer (Y/N): N • Organic N fertilizer (type): n.a. • Synthetic N fertilizer (Y/N): Y • Synthetic N fertilizer (type): urea	• Organic N fertilizer (Y/N): N • Organic N fertilizer (type): n.a. • Synthetic N fertilizer (Y/N): Y • Synthetic N fertilizer (type): urea	• Organic N fertilizer (Y/N): Y • Organic N fertilizer (type): manure • Synthetic N fertilizer (Y/N): Y • Synthetic N fertilizer (type): ammonia	• Organic N fertilizer (Y/N): Y and N • Organic N fertilizer (type): manure • Synthetic N fertilizer (Y/N): Y • Synthetic N fertilizer (type): ammonia or urea
Tillage and/or Residue Management	• Tillage (Y/N): Y • Tillage intensity (type): intensive • Crop residue removal (Y/N): Y	• Tillage (Y/N): Y • Tillage intensity (type): strip • Crop residue removal (Y/N): Y	• Tillage (Y/N): Y • Tillage intensity (type): intensive • Crop residue removal (Y/N): Y	• Tillage (Y/N): Y • Tillage intensity (type): intensive or strip • Crop residue removal (Y/N): Y
Water Management / Irrigation	• Irrigation (Y/N): N • Flooding (Y/N): N	• Irrigation (Y/N): Y • Flooding (Y/N): N	• Irrigation (Y/N): N • Flooding (Y/N): N	• Irrigation (Y/N): Y and N • Flooding (Y/N): N
Grazing Practices	• Grazing (Y/N): N • Grazing system description (continuous, rotational, strip, etc.): n.a. • Animal type (where applicable): n.a. • Harvesting/mowing (Y/N): N	• Grazing (Y/N): N • Grazing system description (continuous, rotational, strip, etc.): n.a. • Animal type (where applicable): n.a. • Harvesting/mowing (Y/N): N	• Grazing (Y/N): N • Grazing system description (continuous, rotational, strip, etc.): n.a. • Animal type (where applicable): n.a. • Harvesting/mowing (Y/N): N	• Grazing (Y/N): N • Grazing system description (continuous, rotational, strip, etc.): n.a. • Animal type (where applicable): n.a. • Harvesting/mowing (Y/N): N
Liming	• Application of calcitic limestone or dolomite (Y/N): Y	• Application of calcitic limestone or dolomite (Y/N): Y	• Application of calcitic limestone or dolomite (Y/N): Y	• Application of calcitic limestone or dolomite (Y/N): Y

⁹ CFG attributes: C4 photosynthetic pathway, annual, no N fixation, herbaceous growth form, non-flooded

¹⁰ CFG attributes: C3 photosynthetic pathway, annual, N fixation, herbaceous growth form, non-flooded

ALM practices	Instance 1	Instance 2	Instance 3	Baseline scenario
Woody biomass management	• Chemical or manual control of competing vegetation (Y/N): n.a. • Burning of vegetation (Y/N): n.a. • Pruning (Y/N): n.a.	• Chemical or manual control of competing vegetation (Y/N): n.a. • Burning of vegetation (Y/N): n.a. • Pruning (Y/N): n.a.	• Chemical or manual control of competing vegetation (Y/N): n.a. • Burning of vegetation (Y/N): n.a. • Pruning (Y/N): n.a.	• Chemical or manual control of competing vegetation (Y/N): n.a. • Burning of vegetation (Y/N): n.a. • Pruning (Y/N): n.a.

In the example, the historical look-back period is set to four years which covers two crop rotations of all instances and the liming practice every 4 years. The column “Baseline scenario of Eligibility Area 1” shows the baseline scenario determined for one eligibility area of a grouped project based on the initial instances 1-3. Instances included in future stages of the project must share the ALM practices summarized in this column. The baseline scenario ALM system (i.e., pre-project management) is described as a yearly crop rotation of maize and soybean, in some instances with one season of wheat cultivation, without cover crops or intercropping, with urea or ammonia fertilization before maize planting, conventional mouldboard tillage or strip tillage before maize planting, crop residue removal every year, no grazing, liming every 4 years, and no woody biomass species.

In grouped projects, the project area may be divided into multiple eligibility areas, where each eligibility area has a common baseline scenario (and additionality demonstration) based upon the initial project activity instances (i.e., management units) at project validation. Any future MUs are added in batches to eligibility areas at future verifications¹¹ and must have baseline ALM practices (as determined through the historical look-back period and Table 6-1 specifications) that fall within the range of variability for each ALM practice established for that eligibility area (i.e., its baseline scenario).

6.2 Development of the Schedule of Activities to Quantify Baseline Emissions

Baseline scenario GHG emissions and C stock changes are determined for each management unit, based on pre-project ALM products and practices (including for livestock and woody biomass) in the specific project context. Baseline scenario practices are set to match the practices implemented in the historical look-back period, creating an annual schedule of activities to be repeated throughout the first baseline period.

¹¹ See the latest version of the VCS Standard for rules on grouped projects, grouped project eligibility areas, and addition of new project activity instances.

Baseline GHG emissions and C stock changes are then modeled by running simulations of the schedule of activities implemented through time (Quantification Approach 1)¹², directly measured in baseline control sites subject to the annual schedule of activities (Quantification Approach 2), and/or calculated using default factors and historical activity data (Quantification Approach 3).

The schedule of activities is determined by assessing practices implemented during the historical look-back period for each management unit. As stated in Section 6.1, the interval over which practices are assessed, x years, must be a minimum of three years and must include at least one complete crop rotation, where applicable. Where a crop rotation is not implemented in the baseline, x must be a minimum of three years. For each year, $t = -1$ to $t = -x$ (i.e., years preceding project start), information on ALM practices must be determined, per the requirements and guidelines presented in Table 6-1, Box 1 and Figure 1.

If the crop rotation is longer than three years, it is permissible to extrapolate management practices to previous years in the crop rotation, as long as management information can be determined according to the Box 1 requirements for at least one season for each crop type. For example, a project may establish a longer historical look-back period (i.e., >3 years) to appropriately capture a complete crop rotation of corn and soy but use detailed management information from just the most recent 3 years of the crops. Where woody biomass is included in the baseline, the schedule of activities must include any management activities affecting that C pool (e.g., periodic removal of competing woody vegetation).

Table 6-1 defines the information requirements for the schedule of activities. If a qualitative variable is recorded as “Yes” during baseline scenario determination, the corresponding quantitative data must be used as an input for the schedule of activities. A “No” response on a qualitative variable in Table 6-1 would mean there is no quantitative data related to that practice.

The schedule of activities must be defined and reported for each management unit to serve as the basis for baseline emission quantification. The schedule of activities, beginning with year t_{-x} , is applied in the baseline scenario, from t_1 onward through the baseline period. Under Quantification Approach 1, the schedule of activities is modeled using either a “fixed” or “blended” baseline as detailed in Section 8.4 and Appendix 9, repeating every x years through the end of the first baseline period. Under Quantification Approach 2, a representative schedule of activities must be developed as detailed in the next section. The representative schedule of activities is then applied in baseline control sites, as detailed in Section 8.4, enabling the direct measurement of baseline SOC stocks.

¹² Note that under Quantification Approach 1, direct SOC stock measurements are also required at time t_0 ($t = 0$) years to serve as model input for modeling the baseline scenario. Per Table 8-1, baseline SOC stocks may be (back-)modeled to t_0 from measurements collected within ± 5 years of t_0 .

6.2.1 Quantification Approach 2: Representative Schedule of Activities for Baseline Control Sites

In contrast to QA1, it is not feasible to determine baseline emissions in all management units under QA2 because the baseline scenario must be directly measured. Therefore, under QA2 baseline SOC stocks are directly measured in baseline control sites matched to multiple management units within one or more quantification units through the similarity criteria listed in Table 6-3 (see Section 8.4 for further details on matching, establishing and managing baseline control sites). Baseline control sites implement a representative schedule of activities covering the most frequently grown crop(s) and associated management practices for each year in the historical look-back period.

The process for developing a representative schedule of activities is as follows:

Step 1: For all management units, determine the management units' specific schedules of activities according to Box 1 and Figure 1.

Step 2: Management units are grouped by the similarity criteria specified in Table 6-3. This results in distinct grouped management units.

Step 3: Identify the most representative sequence of crops and associated practices or grassland management in each management unit group (see example in Appendix 9) to be implemented in each baseline control site. Projects may group by crop functional group (CFG) to perform this analysis.

Step 3a: Determine the representative sequence of crops over the historical lookback period in the groups of management units: Crop sequence is the starting point since, in cropping systems, management choices often stem from crop type grown. For each year of the historical lookback period, determine the crop type grown on the largest area across the grouped management units. For example, the representative crop type in t_3 would be the crop grown on the most hectares in t_3 across the grouped management units. Repeat for all years of the historical lookback period.

Step 3b: Determine the most frequently implemented qualitative management activities for each crop by taking the mode: Using data from all grouped management units that grew the same crop in one or more years of the historical lookback period, determine the associated qualitative management practices most often associated with growing that crop (mode of each qualitative practice).

Step 3c: Determine the central tendency of quantitative management activities for each crop by taking the mean: Using data from all grouped management units that grew the target crop in one or more years of the historical lookback period and employed the mode management activities, calculate the mean quantitative values associated with those qualitative practices that were most often implemented.

Table 6-3. Similarity criteria for linking baseline control sites to QUs under Quantification Approach 2

	Control Site Similarity Criterion	Threshold ^a
Biophysical criteria	Topography	Most frequent slope class ¹³ by area must be the same in the QUs and control sites (to be determined from a slope map or via a GIS slope analysis ¹⁴).
	Soil texture to depth of project boundary (minimum 30 cm)	Most frequent soil texture class by area must be in the same FAO ¹⁵ soil textural class as the most frequent soil texture class by area in the linked QU.
	WRB Reference soil group	Soil group must be within the same first-level reference soil group, according to the <i>World Reference Base for Soil Resources</i> , ¹⁶ as the linked QU.
Biophysical criteria	Historical land cover ^b	For lands converted up to 50 years prior to the project start date, the baseline control site must be converted from the same major land cover type (e.g., forestland, grassland, savanna) as the majority of the area in the linked QU. Conversion of the baseline control site must have occurred within <u>±10 years of the median year of conversion across all or an unbiased, representative sample of management units in the linked QU.</u>
	Native vegetation	The site must be within the same terrestrial ecoregion (Olson 2001 and updated in Dinerstein 2017) as the linked QU <u>or the predominant ecoregion if the QU spans multiple.</u>
	Climate zone	The site must be within the same IPCC-defined climate zone as the linked QU or the predominant climate zone in the QU.
Woody biomass criteria	Pre-existing woody biomass	The average basal area of woody stems > 10 cm in diameter at breast height (DBH) in the control site must be within +/- 20% of the average basal area of woody stems > 10 cm in DBH in the linked QU.
	Advanced regeneration	The average density of woody stems 2.5 – 10 cm in DBH in the control site must be within +/-20% of the average density of woody stems 2.5 – 10 cm in DBH in the linked QU.
	Proximal seed sources	In the direction of prevailing winds, the distance from the centroid of the control site to the closest existing tree line must not exceed the average distance in the linked QU by more than 20%.

¹³ See Appendix 5 for soil slope classifications

¹⁴ See Appendix 5 for workflow steps to determine the most frequent slope class using geographical information systems (GIS)

¹⁵ See Annex 4 in the *FAO World Reference Base for Soil Resources 2014* available at: <https://www.fao.org/3/i3794en/i3794en.pdf> The USDA Soil Texture Calculator (available at: https://www.nrcs.usda.gov/wps/portal/nrcs/detail/soils/survey/?cid=nrcs142p2_054167) may be used to determine the soil texture class based on percent sand and clay content.

¹⁶ See Table 2 of the *FAO World Reference Base for Soil Resources 2014*

	Control Site Similarity Criterion	Threshold ^a
ALM criteria	Historical ALM activities_	Historical ALM activities in the 3+ years prior to initial crediting period start date must match the representative schedule of activities for the linked QU across the following variables: • Tillage (Y/N^d) • Crop Functional Group(s) grown and/or livestock classes (browsers or grazers) • Application of organic amendments (Y/N) Note that not all of these activities will be universally relevant to all agricultural systems. Therefore, projects must provide evidence supporting the selected historical ALM activities used to link control sites with QUs. See Box 1 for guidance on data sources for establishing historical ALM activities.

^aEstimates of these quantitative thresholds in control sites and QUs must be derived from unbiased, representative sampling or mapped datasets. Accuracy must be ensured through adherence to best practices (to be determined by the project and outlined in the monitoring plan – see Section 9.3).

^bEstimated based on historical satellite or aerial imagery or, where imagery is unavailable, confirmed via local expert attestation

^cEstimated based on measurements taken at the closest continuously monitored weather station not exceeding 50 km from the control site or from a synthetic weather station (e.g., PRISM)

^dY/N: Yes/No

^eWhere crop type in the quantification unit of the project area cannot be matched in the baseline control site, a different crop from the same crop functional group may be selected. Crop functional group is defined in VMD0053 as “Broad category of crop species with similar characteristics (e.g., grasses, legumes, non-legume broadleaf species).”

6.3 Woody Biomass Baseline

Where woody biomass is included in the project GHG boundary, the project proponent must determine whether baseline land management practices allow for woody biomass establishment or accumulation. This determination must be informed by agricultural land management activities occurring during the historical lookback period that influence woody biomass carbon stocks (e.g., tillage, burning, mechanical vegetation removal, or planting), as documented in the schedule of activities. Based on this information, project proponents must evaluate the potential for woody biomass accumulation in the baseline scenario by addressing the following questions:

- Are trees or shrubs present in the project area prior to project implementation?
- Has the project area been cleared of forest or woody vegetation within the past five years?
- Does the cultivation system allow woody vegetation to persist without regular clearing?

Where the answer to at least one of the above questions is “yes,” projects must use QA2 to measure and monitor baseline changes in soil organic carbon (SOC) and above- and belowground woody biomass carbon stocks using the same baseline control sites. Baseline

control sites must meet additional similarity criteria relevant to woody biomass accumulation potential (as specified in Table 6-3). This approach ensures alignment between woody biomass and SOC dynamics in baseline quantification. Where the answer to all questions is “no,” it can be assumed that there is no potential for establishment and accumulation of woody biomass in a business-as-usual scenario, and baseline change in woody biomass carbon stocks and stock change is set to zero.

Where baseline woody biomass carbon stock change is set to zero, either QA1 or QA2 may be applied to quantify baseline scenario SOC stocks. When applying QA1, trees and shrubs must be represented in model validation where they are deliberately integrated into the cropping system (e.g., perennial woody crop systems, silvopasture systems, or multistory agroforestry systems). Otherwise, non-cropping wooded areas must be excluded from the project area.

6.4 Baseline Reassessment

The baseline validity period under this methodology is 10 years. The first baseline validity period starts on the initial crediting start date. Subsequent baseline validity periods start on the day after the previous baseline validity period ends. After each baseline validity period, the baseline scenario must be reassessed for the subsequent baseline validity in accordance with the requirements of the most recent version of the *VCS Standard* to ensure they reflect current agricultural production in the region. However, where regional production practices change and/or data becomes available across shorter timeframes, it is recommended to reassess the baseline every five years. The baseline reassessment must also conform with all other requirements for baseline reassessment of the most recent version of the *VCS Standard*.¹⁷ The following requirements apply in addition to the most recent version of the *VCS Standard* when reassessing the baseline:

- a) The schedule of activities in the baseline scenario is valid for the baseline validity period . At the end of each baseline validity period, production of the commercial crop(s) and/or cultivated grassland management in the baseline scenario is re-evaluated. Published regional (sub-national) agricultural production data from within the five years immediately preceding the end of the current baseline validity period must be consulted.
- b) Where there is evidence of continued production of the relevant commercial crop(s) or cultivated grasslands using the same ALM practices (e.g., average tillage frequency, cover crop planting, fertilization practices, grazing management) in the region, the baseline scenario is valid as-is, continuing with the previous schedule of activities.

¹⁷ For example, under VCS Standard v5.0, projects are required to apply the most recent version of the VCS Program rules, including an active methodology, demonstrate regulatory surplus, revise the project description and submit it for validation. The VCS Standard also provides further requirements for the baseline reassessment for agriculture projects.

Where there is no evidence of continued production of the relevant commercial crop(s) or continued grassland management using practices from the most recent baseline validity period, a new schedule of ALM activities (evaluated against common practices in the region) must be developed following the information sources in Box 1 and Figure 1. Information sources must provide sufficient detail to produce the minimum specifications on ALM practices for the baseline scenario as outlined in

- c) .
- d) Where more than one value is derived from analysis of the current regional data (e.g., where a range of application rates is common practice for a certain cropping or grazing system) or where current practices in the region shows higher emissions or lower rates of carbon stock increase than the most recent baseline scenario, the principle of conservativeness must be applied, selecting the value or practice that results in the lowest expected emissions (or highest rate of carbon stock increase) in the baseline scenario.
- e) Where evidence is not management unit-specific, conservatively derived management unit-specific values must be supported by a documented method justifying the appropriateness of selection.
- f) For grazing systems with livestock: Where the regional stocking rate has changed relative to that in the most recent baseline scenario, the new regional stocking rate must be used (even if it is higher than the stocking rate in the most recent baseline scenario) for calculating baseline emissions.

7 ADDITIONALITY

This methodology uses a project method for the demonstration of additionality. For grouped projects, demonstration of additionality must be based upon the initial instances of each eligibility area (see further details in the most recent version of the *VCS Standard*). Where the additionality of the initial project activity instances cannot be demonstrated for the entire eligibility area, the eligibility area must be redefined or divided such that all instances in the area share the same additionality demonstration.

To demonstrate additionality, project proponents using this methodology must:

- 1) demonstrate regulatory surplus;
- 2) identify institutional barriers that would prevent the implementation of a change in pre-existing ALM practices; and
- 3) demonstrate that adoption of proposed individual or combined project activities is not common practice.

Further details on each of these steps are provided below.

Step 1: Regulatory surplus

Project proponents must demonstrate regulatory surplus in accordance with the rules and requirements set out in the most recent version of the *VCS Standard*.

Step 2: Barrier analysis

Project proponents must apply the procedures and requirements in Step 2 of the most recent version of *VT0008 Additionality Assessment*.

Step 3: Demonstrate that adoption of proposed individual or combined project activities is not common practice

Project proponents must determine whether the proposed project is common practice in each region (or eligibility area as defined for grouped projects in the *VCS Standard*) included within the project area. Common practice is defined as greater than 20% adoption.¹⁸ The approach to common practice demonstration outlined in this section follows a two-step process: 1) determine if the project activity has below 20% adoption in the project region, and 2) where the adoption rate is greater than 20% or the required data is not available, project proponents must demonstrate that the activity is not common practice by applying Step 4c of *VT0008*.

The proposed project activity may be implemented as an individual practice (e.g., cessation of tillage) or a combined practice (e.g., conservation agriculture encompassing no-tillage, crop rotations and residue retention). Such combined practices are defined as those implemented on the same field/management unit in the same baseline crop rotation cycle or for at least three years where a crop rotation is not implemented (i.e., aligned with Section 6).

Step 3.1: To determine that an individual or a combined practice is not common practice, the adoption rate of the respective individual practice or combined practice must be demonstrated to be lower than 20% where the practices are implemented.

Evidence of practice adoption rates within the project area must be provided in the form of publicly available information contained in:

- a) agricultural census or other government (e.g., survey) data;
- b) peer-reviewed scientific literature;

¹⁸ Twenty percent is the precedent for a common practice threshold established in Section 18 of the CDM *Methodological tool: Common practice*. Available at: <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

Mathur et al. (2007) and Barnes et al. (2014) confirm that approximately 20% is an appropriate threshold for common practice based on analysis of technology adoption and market penetration using different approaches.

- c) independent research data; or
- d) reports or assessments compiled by industry associations.

To demonstrate common practice, the project area must be stratified to the state or provincial level (or equivalent second-order jurisdiction) in the countries where the project is being developed. Where supporting evidence is unavailable at the state/provincial level (e.g., in developing countries), aggregated data or evidence at a subnational, national, or regional level may be used with justification. Where stratification based on geopolitical boundaries is impractical (e.g., due to lack of data), other forms of stratification, such as major soil types or cropping zones, may be used with justification. The same stratification approach and data sources must be applied across the entire project to maintain the integrity of the common practice demonstration. The common practice assessment data source selected must be justified to be appropriate for the stratified project area.

Where evidence for a proposed project activity in the region is not available from any of these sources, the project proponent may obtain a signed and dated attestation statement from a qualified independent local expert (e.g., agricultural extension agent, accredited agronomist) estimating the adoption rate. All projects using independent local expert attestation must provide the qualifications of the expert and the methods used to inform their analysis. It is recommended that the contracted expert has professional indemnity insurance.

Step 3.2: Where the best available data from the above sources for an individual or combined practice determines that the adoption rate is greater than 20% or the required data is not available, project proponents must demonstrate that the activity (i.e., ALM practice) is not common practice by applying Step 4c of VT0008. This step involves identifying essential distinctions between the proposed project activity and related activities (i.e., ALM practices) that may be captured under Step 3.1. When applying Step 4c of VT0008, the information sources for N_{all} and N_{diff} must cover the same geographic area (i.e., the project region), and the analysis must be on a land area basis (e.g., hectares).

For example, a project intends to introduce cover crops as a project activity in the project region where the adoption rate per the government agricultural census is 22% covering approximately 2 million hectares (N_{all}). Applying the common practice analysis per Step 4c of VT0008, the project identifies the following essential distinction: government subsidies for cover crop seed are accessed by farmers managing 1.7 million hectares (N_{diff}) in the project region, but are not available to the project activity. The calculation of factor F per Step 4c of VT0008 would therefore yield 15% demonstrating that the proposed project activity is not considered common practice.